

Predicting Laptop Prices Using Supervised Machine Learning Models

Dataset

- ❖ Name: Laptop Price Dataset
- ❖ Source: Kaggle
- ❖ Rows: ~1000+
- ❖ Columns: ~13

Source Link

<https://www.kaggle.com/datasets/muhammetvarl/laptop-price>

Target Variable

- Price / Price_euros

Task Type

- Regression

Problem Statement

To predict laptop prices using product features and evaluate the effectiveness of different machine learning models in improving prediction accuracy

Research Questions

1. How well do baseline machine learning models predict laptop prices using key product specifications?
2. How does the performance of advanced regression models compare in predicting laptop prices using key product specifications?
3. What is the impact of different preprocessing strategies on the accuracy of laptop price prediction models?
4. What are the most important features influencing laptop price prediction, and how can their impact be interpreted?
5. How do different evaluation metrics influence the ranking of machine learning models in predicting laptop prices?
6. How robust and generalizable is the model when exposed to variations such as cross-validation, noise, and missing data?
7. Which machine learning model is most suitable for laptop price prediction when considering multiple decision criteria such as performance, interpretability, robustness, and deployment feasibility?

Methodology

- ❖ Data cleaning and preprocessing
- ❖ Feature encoding
- ❖ Train-test split
- ❖ Model training and evaluation

Models Used

- ❖ Linear Regression
- ❖ Decision Tree Regressor
- ❖ K-Nearest Neighbors (KNN)
- ❖ Random Forest Regressor
- ❖ XGBoost Regressor
- ❖ Support Vector Machine (Linear SVR)

Evaluation Metrics

- ❖ MAE (Mean Absolute Error)
- ❖ RMSE (Root Mean Squared Error)
- ❖ R² Score (Coefficient of Determination)
- ❖ Standard Deviation